◇ VisionFlow

Coordination Engineering for Federated Human–AI Intelligence

The open-source coordination harness where diverse intelligence collaborates across trust boundaries at scale. Self-sovereign data. Formal reasoning. Cryptographic provenance. Human-in-the-loop governance.

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5	92	185+	55×	90+	6
Federated Substrates	CUDA Kernels	MCP Tools	GPU Speedup	Agent Skills	OSS Repos

THE PROBLEM

Hard problems share a common shape

Climate modelling, drug discovery, creative production at scale — these require diverse intelligence collaborating across trust boundaries where centralised coordination breaks down.

73% of frontline AI adoption happens without management sign-off.

Uncoordinated agents waste **60–80% of tokens** on context rediscovery, duplicate reasoning, and contradictory outputs.

WHY TOKEN VENDORS SHOULD CARE

Coordination is the token multiplier

- **3–5× effective throughput gain** — shared ontology eliminates re-derivation; provenance eliminates re-validation; governance prevents decision spin
- **10,000:1 problem-to-token ratio** — $2.6B drug compounds justify $10K–100K coordinated reasoning if one wrong conclusion is prevented
- **Break-even at 3 agents** — ontology, identity spine, governance are shared infra, not per-agent costs
- **Model-agnostic** — any LLM behind MCP tools. Multi-vendor consumption scales with problem complexity, not lock-in

ARCHITECTURE

Five substrates, one cryptographic identity spine

Every actor—human, agent, server—shares a `did:nostr` secp256k1 keypair verified at every layer.

VisionClaw | Knowledge Engineering
OWL 2 EL reasoning, 92 CUDA kernels, 55× GPU speedup, 3D graph, 17 MCP tools

Agentbox | Harness Engineering
90+ skills, 185+ MCP tools, governance relay, BIP-340 identity, Nix reproducible

solid-pod-rs | Cryptographic Foundation
Rust Solid server, DID:Nostr + WAC + Web Ledgers, NIP-98 + OIDC + WebAuthn, HTTP 402

nostr-rust-forum | Governance UI
Judgment Broker, Agent Control Surface (kinds 31400–31405), Leptos WASM, passkey auth

DreamLab Edge | Branded Deployment
React SPA + WASM forum, CF Workers edge, operator overlay, live at dreamlab-ai.com

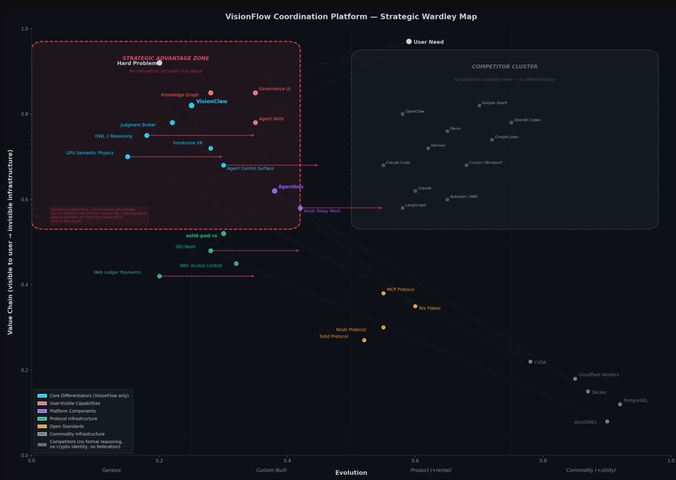
HUMAN-IN-THE-LOOP GOVERNANCE

The Judgment Broker

Agent publishes control panel (kind 31400) → forum renders to human → NIP-98 signed decision → VisionClaw gates knowledge mutations. Every decision is an immutable Nostr event. **90% reduction in audit reconstruction cost.**

STRATEGIC POSITIONING — WARDLEY MAP

Every differentiator sits where no competitor can follow



VisionFlow's core differentiators (red zone, left) occupy the genesis/custom-built quadrant: OWL 2 formal reasoning, GPU semantic physics, the Judgment Broker, cryptographic identity (DID:Nostr), and immersive XR. No competitor has built here. All 11 competing platforms (grey cluster, right) sit in the commodity zone — interchangeable agent frameworks with no structural differentiation. The red dashed boundary marks the **strategic advantage zone**: capabilities that require deep protocol engineering and cannot be replicated by wrapping an API.

TOKEN ECONOMICS

Tokenmaxing for hard problems is rational

Uncoordinated Agents
5–8× token waste
↓
+ Shared Ontology
Eliminate re-derivation
↓
+ Provenance Chain
Eliminate re-validation
↓
+ Judgment Broker
Eliminate decision spin
↓
Coordinated Mesh
3–5× effective throughput

SCALING

Single Operator — One API key, one container, minutes to deploy

Team (50+) — Shared relay mesh, Judgment Broker oversight, validated at scale

Federated Enterprise — Sovereign nodes, trust via did:nostr, horizontal scaling

Open source. Model-agnostic. Federation-ready.
Every token produces auditable, governed, provenance-tracked reasoning.
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